

Second Terminal Examination – 2082

Class : XI

Subject: Physics (1011)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

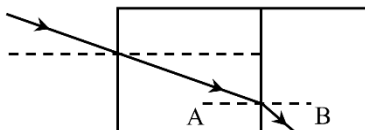
Group A

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

- The refractive index μ of a material is given by relation $\mu = A + \frac{B}{\lambda^2}$, where λ is wavelength of light., then dimensions for the constant B are same as that of
a. Speed b. volume c. distance d. area
- A force of 6 N acts at an angle of 30° to the vertical. The component of force along horizontal direction is
a. 3 N b. $3\sqrt{3}$ N c. $4\sqrt{3}$ N d. $2\sqrt{3}$ N
- If the initial velocity of a projectile is increased by 20%, its horizontal range increases by
a. 20% b. 40% c. 44% d. 50%
- A body is dropped from the top of a tower. The distance covered in the last second of downward motion is 35m. The height of the tower is
a. 45m b. 80m c. 125 m d. 180m
- A shot of mass 20g is projected with a velocity of 200 m/s from a gun of mass 2 Kg. Find the velocity with which the gun recoils.
a. 2 m/s b. -2m/s c. 3m/s d. -3m/s
- A Centigrade and Fahrenheit thermometer are placed in hot water then water is cooled. The fall of temperature in Fahrenheit thermometer, when the Centigrade thermometer read the fall in temperature as 45°C is:
a. 54°F b. 64°F c. 81°F d. 91°F
- A man looks his face on a mirror and observes smaller image. The mirror must be
a. plane mirror b. concave mirror
c. convex mirror d. all of above mirror
- Two spheres A and B of exactly same mass are given equal positive and negative charges respectively their masses after charging
a. remains unaffected b. mass of A > mass of B
c. mass of A < mass of B d. nothing can be said

9. If light travel from A to B in the given figure,

- a. $\mu_A = \mu_B$
- b. $\mu_A > \mu_B$
- c. $\mu_A < \mu_B$
- d. $\mu_A = \mu_B = 0$



10. The no. of electrons contained in one coulomb of charge is
- a. 1.67×10^{31}
 - b. 9.1×10^{-31}
 - c. 6.25×10^{18}
 - d. 6.25×10^{20}
11. The electric field intensity inside the surface of hollow charged sphere is
- a. Zero
 - b. greater than its surface
 - c. smaller than its surface
 - d. equal on every points

Group – B

Give short answer to the following questions.

[8 × 5 = 40]

12. a) A student writes the formula of viscous force as $F = 6\pi\eta rv$, where 'η' is the coefficient of viscosity, 'v' is the terminal velocity and 'r' is the radius of spherical object. Find the dimensions of η. [2]
 b) Convert 19.3 g/cc into kg/m³ using dimensional analysis. [2]
 c) If the displacement (y) is related with time (t) by the relation:
 $y = at + bt^2 + ct^3$
 Find the dimension of 'b'. [1]
13. a) What is rectangular resolution of vectors? [1]
 b) A person throws a ball due west at 6m/s. Wind is blowing along south at 8m/s. Calculate the magnitude and direction of resultant vector. [2]
 c) A force (in Newton) expressed in vector notation as $\vec{F} = 4\hat{i} + 7\hat{j} - 3\hat{k}$ is applied on a body and produces a displacement (in meter) $\vec{D} = 3\hat{i} - 2\hat{j} - 5\hat{k}$ in 4 seconds. Estimate the power. [2]
14. a) Second law of motion is called the real law of motion. Formulate first and third law of motion from Newton's second law of motion. [3]
 b) A 0.0450 Kg golf ball, initially at rest is given a speed of 25 m/s when a club strikes it for 2×10^{-3} sec. Calculate the average force on the ball. [2]
15. "The tendency of solid to change its shape, size and volume in the response of heat is termed as its thermal expansion". Answer the following questions on the basis of given statement.
- a) Define cubical expansivity of solid materials. [1]
 - b) Show that $\alpha = \frac{\gamma}{3}$, where the symbols carry usual meanings. [2]
 - c) A steel wire 8 m long and 4 mm in diameter is fixed to two rigid supports. Calculate the increase in tension when the temperature falls by 10°C. [2]
 $[\alpha_s = 12 \times 10^{-6} K^{-1} \text{ and } Y = 2 \times 10^{11} Nm^{-2}]$

16. The expansion of liquid materials occurs along with the expansion of solid vessel on heating. Answer the following questions based on the expansion of liquid
- Why is real expansion of liquid always greater than its apparent expansion. [1]
 - Show that $\gamma_r = \gamma_a + \gamma_g$, where the symbols carry usual meanings. [2]
 - A glass flask with volume 200 cm^3 is filled to brim with mercury at 20°C . How much mercury overflows when the temperature of the system raised to 100°C . [$\gamma_g = 1.2 \times 10^{-5} \text{ K}^{-1}$ and $\gamma_m = 18 \times 10^{-5} \text{ K}^{-1}$] [2]
17. a) Show that mirror formula also holds good for plane mirror. [2]
 b) An erect image, three times the size of the object is obtained with a concave mirror of radius of curvature 36 cm. What is the position of the object? [3]
18. a) Define critical angle and total internal reflection. [1]
 b) Establish the relation of critical angle with refractive index of the medium. [2]
 c) A microscope is focused on a scratch on the bottom of a beaker. Turpentine is poured into the beaker to a depth of 4 cm, and it is found necessary to raise the microscope through a vertical distance of 1.25 cm to bring the scratch again into focus. Find the refractive index of turpentine. [2]
19. a) Define action of point. The top of the Dharahara is made pointed, why? [2]
 b) What is the limitation of Coulombs laws of electrostatics? [1]
 c) How many equal electrons must be placed on each of two small spheres placed 3 cm apart for the force of repulsion between the spheres to be $2.56 \times 10^{-19} \text{ N}$? [2]

OR

- Define the electric field intensity and find the electric field intensity at a point due to plane sheet charged conductor. [1+2]
- An electron of mass $9.1 \times 10^{-31} \text{ kg}$ and charge $1.6 \times 10^{-19} \text{ C}$ is situated in a uniform electric field intensity of $1.2 \times 10^{-4} \text{ V/m}$. Find the time to travel 1 cm from rest. [2]

Group – C

Give long answer to the following questions.

[3 × 8 = 24]

20. a) Describe the types of velocities in projectile motion. [1]
 b) For a projectile projected horizontally from height 'H' above the ground, show that its trajectory is parabolic. [2]
 c) A stone on the edge of a vertical cliff is kicked so that its initial velocity is 9 m/s horizontally. If the cliff is 200 m high. Calculate: [2]
 i) Time taken by stone to reach the ground
 ii) How far from the cliff the stone will hit the ground?
 d) If you drop a ball of certain mass from a height 'h' and at the same instant another ball of same mass is thrown horizontally with a velocity 'u', Which ball will hit the ground first? [2]

- e) Can a body have a zero velocity and still be accelerating? [1]

OR

- a) Using velocity-time graph, derive the expression $s = ut + \frac{1}{2}at^2$, where the symbols carry their usual meaning. [2]

- b) From the v-t graph:

i) Find the acceleration from

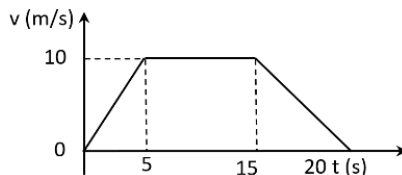
0 to 5 sec.

[1]

ii) Find the displacement from

0 to 20 sec.

[1]



- c) A man walking on a road with a speed of 5km/hr, encounters rainfall vertically with a velocity of 12 km/hr. What is the velocity of rain relative to the man? [3]
- d) If the displacement of a body is proportional to the square of time. State the nature of the motion of the body. [1]
21. Newton's Law of cooling is an experimental technique for the determination of specific heat capacity of unknown liquid.
- a) Draw proper calorimeters arrangement and state the principle used. [1]
- b) Express the mathematical relation for the rate of heat loss by each calorimeter with contents. [2]
- c) Based on the expression of rate of heat loss, deduce the equation of specific heat capacity of liquid. [2]
- d) What is specific heat capacity if the thermal process takes place at constant temperature? [1]
- e) A substance takes 3 minutes in cooling from 50°C to 45°C and 5 minutes in cooling from 45°C to 40°C. Calculate the temperature of surrounding. [2]
22. a) Explain how can you charge a body negatively by the method of electrostatic induction. [3]
- b) A comb drawn through person's hair on a dry day causes 10^{22} electrons to leave the persons hair and stick to the comb. [1+1]
- i) Calculate the charge carried by the comb.
- ii) What will be the increase in mass of comb? ($m_e = 9.1 \times 10^{-31}\text{kg}$)
- c) ABC is an equilateral Δ of side 5 cm. The charge of $-5 \mu\text{C}$ and $+10 \mu\text{C}$ are placed at A and B. Calculate the magnitude and direction of electric field intensity at C. ($\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{Nm}^2/\text{C}^2$) [3]

— The End —